124102019

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
End Sem., Question Paper

Month and year: **May, 2025**
Program: **B. Tech**
Subject: **Programming using Python**
Maximum Marks: **50**

Semester: **2nd**
Course code: **CSIC104**
Time allowed: **03 hrs**

Note: Attempt all questions. All questions carry equal marks.

Q1: a. Discuss the main data types in Python Language. Demonstrate the concept and use of immutable variables in Python with code.

b. Given a text file file.txt that contains a list of phone numbers (one per line), write python program to print all valid phone numbers. You may assume that a valid phone number must appear in one of the following two formats: (xxx) xxx-xxxx or xxx-xxx-xxxx. Where, x means a digit between 0 to 9. Assume that each line in the text file must not contain leading or trailing white spaces.

Example: Assume that file.txt has the following content:
987-123-4567
123 456 7890
(123) 456-7890
The output the following valid phone numbers:
987-123-4567
(123) 456-7890

Q-2: You are given a dictionary itemList = {"Notebook": 50, "Marker": 25, "Sharpener": 10, "Compass": 40}, representing items and their prices.

a) Write a function that accepts this dictionary along with the name of an item and returns the price of that item. If the item does not exist in the dictionary, then it should return -1. For example, if the name of the item is "Compass" then the function should return 40.
b) Write another function to update the dictionary. The function should accept the dictionary, name of an item, and its new price. The function should return the updated dictionary as per the following conditions:

Case 1: If the new price is negative or zero and the item exists in the dictionary, then the item should be removed from the dictionary.
Case 2: If the new price is positive and the item exists in the dictionary, then the price of that item should be updated to the new price.
Case 3: If the new price is positive and the item does not exist in the dictionary, then the item-price pair should be added to the dictionary.

OR

Write a function *Invert_dict()* that takes in a dictionary with immutable values and returns the inverse of the dictionary. The inverse of a dictionary d is another dictionary whose keys are the unique dictionary values in d. The value for a key in the inverse dictionary is a sorted list (increasing order) of all keys in d that have the same value in d.
Example- If d = {1:10, 2:20, 3:30} then *Invert_dict* (d) returns {10: [1], 20: [2], 30: [3]}

If d = {1:10, 2:20, 3:30, 4:30} then *Invert_dict* (d) returns {10: [1], 20: [2], 30: [3, 4]}

Q-3: a) The information of the students of a university is given as the list of tuples. The first entry of the tuple is the name of the student, the second entry is the name of the branch and third entry is the percentage marks he/she secured. Write a python program to perform the following operations:

(i) Sort the branches in non-decreasing order based on average marks in that branch.

(ii) For each branch (already sorted) print the details of the students in non-decreasing order based on their individual score.

b) Write Python classes DailyWage and MonthlyAttendance those contain an employee's daily wage and monthly attendance, respectively. Calculate the employee's monthly salary by multiplying objects of both the classes using operator overloading.

Q-4: a) Write a python program (using TKinter module) that creates a simple calculator application with buttons for arithmetic operations (addition, subtraction, multiplication, division) and displays the result in a label widget. Include error handling for invalid input. Also draw the GUI you have created.

b) A robot moves in a plane starting from the original point (0,0). The robot can move toward UP, DOWN, LEFT and RIGHT with a given step. The trace of robot movement is shown as the following: UP 5 DOWN 3 LEFT 3 RIGHT 2. The numbers after the direction are steps. Write a program to compute the distance from current position after a sequence of movement and original point. If the distance is a float, then just print the nearest integer.

Example: If the following tuples are given as input to the program:

UP 5, DOWN 3, LEFT 3, RIGHT 2

The output of the program should be: 2

Q-5: Consider a class named *Professor* having following features: name, date of birth (Dob), date of joining (DoJ). Create a list of 20 subjects as subject choice. After taking the choice from each professor, allot the subject on the basis of DoJ (preference is given in First come First serve order), in case of tie use DoB, the older guy has higher priority. Write a python program to overload the corresponding comparison operators to decide the priority among professors and print the allotment details.